

# Amitabh Choudhury

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personal-portfolio-cyan-eight-50.vercel.app

## Summary

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Data Engineer with 2 years at Caterpillar — recently shipped a production LLM-powered Anomaly Detection Platform with Cortex Analyst and a three-tier approval workflow. Builds and evaluates GenAI systems outside work: Text-to-SQL with adversarial benchmarking, a hybrid RAG pipeline with RAGAS eval gates, and a quantile MLOps forecasting platform. B.Tech AI/ML (CGPA 8.55).

## Work Experience

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### Caterpillar Inc. Bengaluru, India

*Data Engineer Jun 2025 – Present*

*Promoted from Systems Data Support Analyst within 12 months.*

- Architected and shipped a production LLM-powered Anomaly Detection and Alerting Platform (Streamlit-in-Snowflake) for the PSLD supply chain division; three-tier (L1/L2/L3) rule approval workflow with **Cortex Analyst-generated SQL**, role-based access control, and a self-service rule management interface — the first production GenAI application in the division, cutting anomaly rule deployment from a multi-day manual process to same-day self-service.
- Built a Snowflake-native Access Administration Dashboard for managing L1/L2/L3 reviewer roles across the platform; real-time role resolution, soft-update audit trail with change attribution and reason capture; replatformed the reviewer email notification pipeline from Dataiku to a native Snowflake trigger, eliminating an external tool dependency.
- Led the MineStar MSU SAS-to-Snowflake migration end-to-end: refactored 2,025 lines of legacy SAS into 32 optimised Snowflake objects (69% table reduction via CTE consolidation) across 5 data domains covering ~5,500 global parts; verified cutover with row-level parity reconciliation and retired the legacy mainframe-FTP job.
- Own 25+ production ETL/ELT pipelines (SnapLogic, dbt, Snowflake) with refresh SLAs as tight as 20 minutes, powering Power BI dashboards that drive daily shipment-allocation decisions for ~75 supply-chain stakeholders.
- Designed a Goods Receipt dbt pipeline with SCD Type-2 history at supplier/PO grain; 13 schema and relationship tests enforce data contracts; Snowflake Dynamic Tables auto-refresh summary aggregates every 30 minutes.
- Audited an internal Snowflake Cortex NL-to-SQL prototype and identified critical safety gaps (no write protection, no adversarial eval, no hallucination detection); designed and open-sourced a hardened rebuild with AST-based guardrails and semantic similarity hallucination detection — see Text-to-SQL project.

### *Systems Data Support Analyst Jun 2024 – Jun 2025*

- Supported production Snowflake/SnapLogic/Tidal pipelines for warehouse, procurement, and finance stakeholders; owned data-quality triage and ad hoc analytical SQL feeding procurement-planning decisions.
- Initiated a multi-year warehouse parts usage analysis (Python, pandas) to surface seasonal demand patterns; built a Random Forest forecasting prototype with time-based train/test split — findings and methodology carried forward into the Data Engineer role.

### *Software Engineer Intern Jun 2023 – Sep 2023*

- Implemented KS-test + PSI drift detection on 3 production Tidal data streams; tuned alert thresholds against 6 months of historical data, reducing false-positive alerts by ~30% and cutting unnecessary pipeline re-runs.

## Skills

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**AI / GenAI:** LLM application development, RAG (Retrieval-Augmented Generation), prompt engineering, hallucination detection, embeddings, semantic search, vector databases (Qdrant), BM25/RRF hybrid search, evaluation/evals, RAGAS, Langfuse, Claude Sonnet API, sentence-transformers (BGE-large)

**ML / MLOps:** LightGBM (LambdaMART, Quantile), MLflow, Evidently AI, scikit-learn, statistics (KS-test, PSI, quantile regression), time series forecasting, FastAPI

**Data Engineering:** ETL/ELT, data pipeline orchestration, data modeling, Snowflake (Cortex, Dynamic Tables, Tasks, Streams), dbt (SCD Type-2, snapshots, tests), Apache Iceberg, DuckDB, Dagster, Soda Core, SnapLogic, Tidal

**Languages:** Python, SQL, C++

**Analytics & BI:** Pandas, Power BI

**Infra & Dev:** Docker, GitHub Actions CI/CD, Git, PostgreSQL 15

## Projects

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## Text-to-SQL with Guardrails & Hallucination Detection | [GitHub](#)

*FastAPI · PostgreSQL 15 · Claude Sonnet · sqlparse · sentence-transformers · Docker*

- NL-to-SQL over a supply chain PostgreSQL database; sqlparse AST guardrails (token-type checking, not regex) hard-block DDL/DML, injection, and catalog access; all execution runs inside PostgreSQL SET TRANSACTION READ ONLY as a database-enforced fallback. **100% guardrail block rate on 10 seeded injection tests; zero unsafe queries.**
- 50-question adversarial benchmark (JOINS, CTEs, date arithmetic, ranking, 6 unanswerable queries); back-translation hallucination detection via MiniLM-L6 cosine similarity; multi-query validation surfaces SQL divergence to the user; 6 Architecture Decision Records.

## Supply Chain Demand Forecasting MLOps | [GitHub](#)

*LightGBM · MLflow · Evidently AI · FastAPI · Streamlit · Docker · GitHub Actions*

- End-to-end forecasting platform for 500 spare-parts SKUs rebuilt from a Caterpillar prototype; QuantileLightGBM (P10/P50/P90) via walk-forward CV achieves **34.9% WAPE and 82.6% interval coverage** — 46% over seasonal-naive best baseline; versioned Parquet feature store with 33 hand-crafted features (feature engineering: lag, rolling stats, seasonality, stockout-adjusted).
- MLflow model registry with **promotion gate** (candidate must beat production WAPE by >2%); Evidently AI separates data drift and concept drift with two automated responses; GitHub Actions CI/CD pipeline triggers automated retraining on drift alerts; FastAPI model serving returns P10/P50/P90 with model-version headers.

## Hybrid RAG System with Evaluation Suite | [GitHub](#) | [Live Demo](#)

*BGE-large · BM25/RRF · Qdrant · RAGAS · Claude Sonnet · Langfuse · GitHub Actions · Docker · Render*

- Production RAG pipeline over Snowflake and dbt documentation; dense retrieval (BGE-large, Qdrant) fused with BM25 via RRF (k=60); BGE cross-encoder reranker; Claude Haiku query rewriter; answers grounded with cited source chunks; live on Render.
- 50-question adversarial benchmark across 4 tiers; best configuration achieves **0.86 RAGAS faithfulness** on conceptual queries vs. 0.31 baseline — 3-config ablation quantifies reranker vs. hybrid retrieval contribution; CI eval gate fails build on faithfulness regression; all retrievals traced in Langfuse.

## GitHub Archive Lakehouse | [GitHub](#)

*Apache Iceberg · DuckDB · Dagster · Soda Core · MinIO · Streamlit · Docker*

- Bronze/Silver/Gold Iceberg lakehouse ingesting **7.4M+ GitHub Archive events**; manifest-tracked crash-resumable pipeline; Silver deduplicates by event ID, parses 15 event-type payloads, routes ~600 malformed rows to quarantine with error reason.
- **36 Soda Core data quality checks** gate each Dagster run; GitHub Actions CI/CD pipeline fails on >3 check failures; Software-Defined Assets provide full lineage; Iceberg time-travel demonstrated on a 3.8M-row table; 5 documented failure scenarios with exact recovery commands.

## Achievements

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- **LeetCode Knight** — max rating 1923; global rank 283/29,865 in Weekly Contest 366 (top 1%) | amitabhchoudhury19999
- **Codeforces** — max rating 1800 (Expert) | amitabh1208
- **CodeChef** — max rating 1711 (4★) | amitabh1208
- **Centurion Hackathon (National Level)** — Team Lead; 13th / 500+ participants.

## Education

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**M.S. Ramaiah University of Applied Sciences, Bengaluru** — B.Tech in Artificial Intelligence and Machine Learning, CGPA 8.55/10 2020 – Apr 2024